

Twelve Unit Cubes in a Cube of Side 2.9315, Found by Hardening Balls into Cubes

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[TODO: date]

Abstract

We give a packing of 12 unit cubes in a cube of side 2.9315185, with every pair and every wall at least 10^{-6} apart, improving the previous best known side 2.9327717687 (H. Lin, July 2026 [8]). The packing was found by a soft-to-rigid homotopy: unit-diameter balls are compressed in a shrinking box and then deformed through rounded cubes into rigid unit cubes while inward pressure on the box continues. The final configuration is tightened with a constrained solver that keeps every corner inside the box and a separating plane between every nearby pair, and validated by an exact rational certificate. On unit squares in a square, the same method recovers the best known packings for $n = 11$ (now proved optimal), 17, 18, 19, 26 and 27, and outperforms a rigid-start control on the same seeds. We report where the method fails ($n = 28, 29$ squares; $n = 10, 11$ cubes so far), an ablation showing that an oriented intermediate shape does not help, and code to reproduce every result.

1 Introduction

Let $s_d(n)$ be the side of the smallest d -dimensional cube that contains n non-overlapping unit cubes, free to rotate. For squares ($d = 2$) the problem has a long record history [1, 7, 3, 9]; the smallest unsettled case, $n = 11$, was recently closed [9]. For cubes ($d = 3$) far fewer results exist. Friedman’s catalogue [2] lists non-trivial packings for $n = 9$ –14 and 28–33; the entries for $n = 11$ and 12 were improved in 2026 (T. Berthold et al. in March for $n = 11$; H. Lin in July for $n = 11$ and 12, with coordinates published in [8]), and for every other $n < 34$ the trivial packing is the best known.

Contribution.

1. A packing of 12 unit cubes in a cube of side 2.9315185094797 with every pair and every wall separated by at least 10^{-6} (§3), 0.0012533 below the previous record 2.9327717687048653. It is certified in exact rational arithmetic.
2. A simple search method, soft-to-rigid homotopy with inward pressure (§2), and an exact tightening step that turns its output into a locally optimal packing to 10^{-12} .
3. An empirical record of what the method does and does not reach: in 2D it recovers six published records from random starts and beats a rigid-start control; it fails on $n = 28, 29$ squares; an oriented intermediate shape does not raise its hit rate (§4, §5).

This is a search result, not an optimality proof.

*[TODO: affiliation and contact]. Experiments, code and verification were developed with the assistance of Claude (Anthropic); see the acknowledgements.

Related methods. Gensane and Ryckelynck’s perturbation–compression algorithm found dense square packings, including $n = 11$, from random starts [4]. Recent square records ($n = 28, 29, 39, 41, 50, 51, 55$) come from rigid simulated annealing and exact refinement [9]. Our method differs in starting from balls and changing the shapes, not just their positions, during the search.

2 Method

Shape family. A *rounded cube* of softness $r \in [0, \frac{1}{2}]$ is a cube of half-side $\frac{1}{2} - r$ dilated by a ball of radius r (the Minkowski sum). At $r = \frac{1}{2}$ it is a unit-diameter ball, at $r = 0$ a unit cube, and every member lies inside the unit cube with the same centre and orientation. Hardening ($r \downarrow 0$) therefore only grows the shapes, and the container must expand to accommodate them. The square version uses rounded squares.

Energy and pressure. For centres c_i , orientations q_i (quaternions in 3D, angles in 2D) and container side L ,

$$E = \sum_{i < j} (2r - \text{sd}(i, j))_+^2 + \sum_i \sum_{\text{walls}} (\text{violation})_+^2 + \mu L,$$

where $\text{sd}(i, j)$ is the signed distance between the cores (cube of half-side $\frac{1}{2} - r$): exact Euclidean distance when separated (vertex–face and edge–edge features), and minus the separating-axis penetration depth when overlapping (15 axes for cubes [5]). The term μL is inward pressure; the box resizes symmetrically so all walls push. Positions, orientations and L follow momentum gradient descent with analytic gradients (checked against finite differences on 72,000 random cube-pair gradient components and 120,000 square-pair components).

Schedule. Compress balls under pressure 3μ (5,000 steps); morph $r : \frac{1}{2} \rightarrow 0$ linearly at pressure $\mu = 0.02$ (45,000 steps); settle rigid cubes while μ decays to $2 \cdot 10^{-6}$ (3,000 steps). Optional *annealing* adds Gaussian kicks to positions and angles that decay to zero during compression and morphing. The *rigid control* runs the identical schedule with $r = 0$ throughout.

Legalization. Penalty pressure tolerates small overlaps. Each run is legalized by scaling centres apart about their mean (bisection to the smallest factor at which no pair overlaps by more than 10^{-12} under the separating-axis test) and measuring the tight bounding cube. All sides reported below are legal packings.

Exact tightening. Runs within a threshold of the catalogue are passed to a constrained solver: minimize L over centres, rotations and per-pair separating planes $u_{ij} \cdot p = c_{ij}$, subject to every corner of every cube lying in $[0, L]^3$ and the corners of i and j lying on opposite sides of their plane. We use SLSQP [6] in outer iterations with a trust region, rebuilding the pair list and re-basing rotations each iteration, and accept a step only if the independently legalized side decreases. From near-miss runs the solver converges to Trump’s $s_2(11)$ [10] and Bidwell’s $s_2(17)$ [3] to 10^{-12} , and from a perturbed copy to Friedman’s $s_3(9) = 2 + 1/\sqrt{2}$.

Exact certificate. For a reported 3D packing we expand centres by a factor $1 + 10^{-6}$ about their mean and inflate each half-side to $\frac{1}{2}(1 + 10^{-12})$, which covers floating-point non-orthogonality of the axes. We convert every coordinate to an exact rational, compute all corners exactly, and check (i) every corner lies in $[0, S]^3$ for a rational S and (ii) every pair is strictly separated by an explicit rational plane. A linear program proposes each plane; the check itself uses only exact arithmetic.

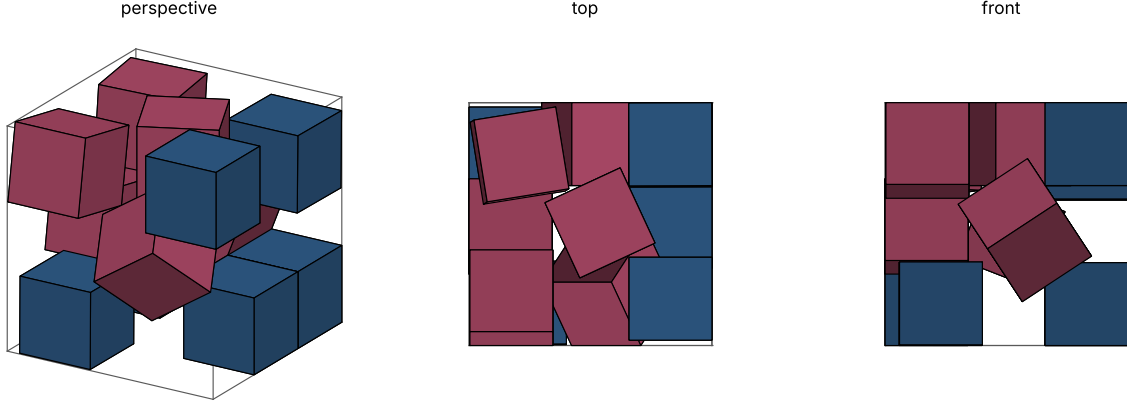


Figure 1: Twelve unit cubes in a cube of side 2.931515. Blue: tilt below 1.5° ; red: tilted.

3 Twelve cubes

Theorem 1. *Twelve unit cubes fit in a cube of side 2.9315185094797 with pairwise and wall clearance at least 10^{-6} . Hence $s_3(12) \leq 2.9315185094797 < 2.9327717687048653$.*

Certificate. The claim file (`claims/cubincub_n12/cubincub_n12.json`, Table 3) lists one pose $[x, y, z, q_w, q_x, q_y, q_z]$ per cube, the format of the previous record file [8]. It was obtained from the tightened packing (side 2.931514577965, with touching contacts) by spreading centres by a factor $1 + 1.0 \cdot 10^{-6}$ until every pair is separated by 10^{-6} , adding a 10^{-6} wall gap and rounding the side up. Two checks pass. (i) A floating-point 15-axis separating-axis test (`verify.py`, 30 lines): minimum wall gap $1.000000000 \cdot 10^{-6}$, minimum pair gap $1.000000001 \cdot 10^{-6}$. (ii) An exact check (`certify_exact.py`): every number is converted to a rational, rotations are the exactly orthogonal $H(q)/|q|^2$, every corner lies in $[0, s]^3$ and every pair is strictly separated by an explicit rational plane.

Four cubes are exactly axis-aligned, two are tilted by less than 1° , and six are tilted by 9° or more (Figure 1). Several quaternions have repeated components (e.g. cube 1: $(0.58389, -0.58389, -0.39884, -0.39884)$), suggesting a closed form that we have not derived. [TODO: attempt exact algebraic value from the active contact graph.]

How it was found. Seed 52 of the first 64 ball→cube starts with annealing ended at legalized side 2.931821; exact tightening lowered it to 2.931514577965. A second start (seed 83) ended at 2.931910 and tightens to 2.931619541: a distinct local optimum, also below the catalogue. Two of the first 128 starts land below 2.93277. The rigid control’s best of 32 starts is 2.99257.

Other cube counts. Table 1 summarizes the first 3D screen. The method matches Friedman’s $n = 9$ packing often, settles in worse arrangements at $n = 10$ and 11, and is the only setup that comes near the catalogue at $n = 12$.

4 Validation on squares

We developed the method on unit squares, where records are dense and recently audited [9, 3].

n	catalogue	starts (ball / rigid)	best ball→cube	best rigid	matches (ball / rigid)
9	2.70711	64 / 32	2.707107	2.707107	16 / 5
10	2.70711	64 / 32	2.767767	2.894428	0 / 0
11	2.88295	64 / 32	2.894427	2.918992	0 / 0
12	2.93277	128 / 32	2.931515	2.992573	—
13	2.956	[TODO:]			
14	2.98995	[TODO:]			

Table 1: First 3D screen (annealed ball→cube vs. rigid control, 53,000 steps per start; best after exact tightening). The best $n = 10$ run is within $2 \cdot 10^{-7}$ of $1 + 5\sqrt{2}/4$; the tightened best $n = 11$ run equals $2 + 2/\sqrt{5} \approx 2.894427$ to 10^{-9} , below the March 2026 value but above Lin’s 2.8829529.

$n = 11$. With annealing and the long schedule, 6 of 160 random starts reach Trump’s packing (tilt 40.18°); 0 of 120 rigid-control starts do (best 3.8877, median 4.000). Trump’s packing has since been proved optimal [9], so this is a sanity check.

$n = 2$ –30 **at a fixed schedule**. Among cases whose best known packing has tilted squares, the method matches $n = 5, 10, 11$ and none from 17 upward at the standard step budget; the rigid control matches 5 and 10. Grid cases split by slack: when the grid has empty cells ($n = 6, 7, 12, 13, 20, 21$) rigid starts fall into it more often (46–100% vs. 2–19%), while for full or nearly full grids ($n = 9, 15, 16, 22$ –25, 30) rigid starts jam at tilts they cannot rotate out of (e.g. $n = 25$: rigid 0/24, balls 7/48).

n	best known (finder)	recovered to	hit rate
17	4.675530 (Bidwell 1998)	10^{-12}	10/960 blob starts; 0/80 rigid
18	4.822876 (Hämäläinen 1980)	10^{-5}	2/24
19	4.885618 (Wainwright 1979)	10^{-12}	2/960
26	5.621320 (Friedman 1997)	10^{-12}	2.5% / 6.3% / 10.4% at $3\times/5\times/10\times$ steps
27	5.707107 (Göbel 1979)	10^{-12}	1/416
28	5.824445 (Ellsworth 2025)	—	best 5.844498
29	5.933833 (Schadt 2025)	—	best 5.938996

Table 2: Squares: best known packings reached from random starts. No run went below a catalogue value in about 5,000 starts at $n \geq 17$.

Old records recovered with more steps and exact tightening. Exact tightening matters: raw runs stop about 10^{-4} above the record because of penalty slack, and the solver also pulls runs from up to 0.022 away into the record basin ($n = 27$). After tightening, every arrangement within 0.001 of the catalogue at $n = 17$ and 19 was the catalogue packing itself.

5 Ablations and negative results

Oriented intermediate shape. Disks carry no orientation, and on the $n = 11$ hit the angles lock in only once $r < 0.06$. We tested morphing through regular octagons cut back to squares, which exert torque from the start. On the same seeds and steps ($n = 17, 19$, seeds 1–160), octagons hit 0/320 vs. 4/320 for annealed disks (one-sided $p \approx 0.06$), with no better medians and $3\times$ the cost. Late orientation commitment appears to help rather than hurt.

Penalty polish. A fixed-box squeeze-and-shake stage changed sides by at most 0.009 and never changed the arrangement; it is superseded by exact tightening.

Scaling. Hit rates fall with n at a fixed budget. At $n = 26$ they rise with hardening time; at $n = 27$ – 29 they do not.

6 Discussion

The method’s value is the basin it lands in: starting from balls, configurations are dense before orientation matters, and rotation is decided only as corners appear. For 3D, where the catalogue is sparse, this was enough to improve $n = 12$ within 128 starts. [TODO: Update with $n=11, 13, 14$ and larger 3D screens.]

Limitations. (i) No optimality claims. (ii) The catalogue values for cubes were read from Friedman’s page images, and Lin’s packings are not described in text; 2.93277 is truncated (“2.93277+”), which does not affect a 0.00126 margin. (iii) Hit rates are from modest sample sizes and are reported with counts.

Reproducibility. All code, seeds, run logs and certificates are at <https://github.com/yoheinakajima/soft-to-rigid-packing>. `scripts/reproduce_n12.sh` regenerates the $n = 12$ packing and its certificate in a few minutes on a laptop.

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A Coordinates for $n = 12$

Centres and unit quaternions (w, x, y, z) ; cube i occupies $c_i + R(q_i)[- \frac{1}{2}, \frac{1}{2}]^3$ in the box $[0, 2.9315185094797]^3$. Full-precision values: `claims/cubincub_n12/cubincub_n12.json`.

Table 3: Coordinates of the $n = 12$ packing.

i	x	y	z	w	q_x	q_y	q_z
1	1.5243900087	2.4315173049	1.3320181026	0.5838886869	-0.5838886869	-0.3988408220	-0.3988408220
2	2.4315175095	2.4315173049	0.5000010000	0.7071067812	0.7071067812	-0.0000000000	0.0000000000
3	0.5000010000	1.4369556928	1.4377218077	0.5395289954	0.4570650535	0.4570650535	-0.5395289954
4	0.5098745912	0.5769319953	2.3545775829	0.4560511351	-0.5403863020	-0.4560472394	0.5403896022
5	0.6086104699	2.2929309153	2.3887769910	0.0258332833	0.7609821302	-0.6482073268	-0.0081302199
6	2.4315175095	1.4315163049	0.5000010000	0.0000000000	-0.0000000000	-0.0000000000	-1.0000000000
7	2.4298431049	2.4215276465	2.2620542416	0.0000269579	0.0016621483	-0.9999712786	0.0073944948
8	1.5800409188	1.4806927085	2.4315175095	0.0000000000	-0.9765112947	-0.2154662186	-0.0000000000
9	1.6870691741	0.7046058868	1.3931640729	0.1630483719	-0.8266039691	-0.3023204668	0.4458065074
10	0.6714203714	0.5125513149	0.5042794969	0.0019279356	0.0017767559	-0.7100527881	-0.7041435680
11	2.4315175095	0.5648571561	2.4315175095	0.5000000000	-0.5000000000	-0.5000000000	0.5000000000
12	0.5000010000	2.3777515563	0.5000010000	0.7071067812	-0.7071067812	0.0000000000	0.0000000000